

**THE LIVER SYSTEM
IN TRADITIONAL CHINESE MEDICINE**
Dr. Yun Xiao
Bastyr University

Liver is located in the right hypochondriac region beneath the diaphragm. TCM holds that Liver houses the soul; stores blood, and dominates tendons. In the five elements theory, Liver belongs to the element wood, in which nature is active and ascending. It is a strong and rigid organ. This is why ancient Chinese medicine also sees the entire human body as an empire with the Heart as its emperor and the Liver as the general of its army. The main physiological functions of the Liver are maintains the free flow of Qi, and storing Blood. Other associated functions of the Liver include dominates the tendons, manifests in the nails, and opening into the eyes. Liver also represents anger among the emotions and making the tear as the Liver's body fluid. The gallbladder is located beneath the right lobe of the liver. The Liver channel connects with the gallbladder to form an external-internal relationship.

The Pattern Identification and Therapeutics of the Liver Disorders

Liver disorders are generally classified into two categories: deficiency and excess.

The deficiency patterns include: deficiency of Liver blood and insufficiency of Liver Yin.

The excess patterns include: Stagnation of Liver Qi, Upflaming (inflaming) of Liver Fire, pathogenic cold, damp heat, or phlegm accumulating in the Liver, Gallbladder, and their channels. The others, such as hyperactivity of Liver Yang and stirring of Liver Wind, belong to intermingled deficiency and excess.

The common symptoms and signs of the Liver disorders may be seen as: distension and pain in the hypochondriac region, depression, irritability, anger, dizziness, vertigo, headache, tremor, convulsion, eye disorders, abnormal menstruation, bitter taste in the mouth, jaundice, palpitation and insomnia, etc.

Hence, each pattern of the liver disorders will be discussed systematically and respectively in the following text.

1. Stagnation of Liver Qi

This is one of the very common patterns in clinical practice. It is due to failure of the Liver to unobstruct and diffuse Qi. The excessive and harmful Qi is depressed and stagnated in the Liver. It usually results from emotional disharmony such as depression, frustration, irritability, and anger, or the other pathogenic factors that attack the Liver.

Clinical Manifestation:

1. Mentally - depression and easy to lose temper.
2. Physically – wandering and migrating pain in hypochondriac region.
3. Energetically – wiry pulse most remarkably in the Left Hand.

Emotional depression
Frustration
Irritability
Anger
Unhappy
Stressful

Hypochondriac pain
Discomfort in the chest
Sighing
Sensation of "plum pit Qi" in the throat

Anorexia
Poor digestion
Nausea
Belching
Sour regurgitation and vomiting

Epigastric pain and distension
Borborygmus
Loose stools

Jaundice
Bitter taste

Male Premature-ejaculating & ED
Female PMS
Distention and soreness of the lower abdomen & breasts
Irregular menstruation
Amenorrhea
Infertility
Decreased libido

Pulse is wiry
Tongue is normal or purplish

Prognosis & Complications

Liver Heat
Liver Fire
Five Elements Relationships

Differentiation

Liver Qi vs. Liver Heat vs. Liver Fire
Liver Qi Stagnation vs. Heart Shen Disturbed

Therapeutics:

The treatment principle is to invigorate Liver Qi and disperse stagnation.

Acupuncture:

Tai Chong	LV 3
Wai Guan	SJ 5
He Gu	LI 4
Yanglingquan	GB 34
Qi Men	LV 14

Herbal Formulation:

Xiao Yao San 逍遥散

- Chai Hu (Bupleurum) 9gm
- Dang Gui (angelica) 9gm
- Bai Shao (White peony) 9gm
- Bai Zhu (Astractylodes) 9gm
- Fu Ling (poria) 12gm
- Sheng Jiang (ginger) 15gm
- Bo He (ment) 6gm
- Zhi Gan Cao (Licorice) 6gm

2. Upflaming of Liver Fire

This is the manifestation of the rebelling of Liver Qi and Fire, which are caused by emotional disturbance. The depression of Liver Qi transforms into the Liver Fire; or pathogenic Heat accumulates in the internal body.

Clinical Manifestation:

1. Headache
2. Anger, violent
3. Redness:
 - a. Red eyes
 - b. Red face

Anger
Anxiety
Panic
Nightmares
Violent behavior
Insomnia

Hypochondriac pain
Headache
Eye distension
Redness:
Red eyes
Red face

Thirsty
Dry mouth
Inner ear pain, discharge pus (Otitis media)

Cravings of alcohol, sugar, smoking
Eating disorders can come from these areas, can cause regurgitation
Constipation
Hematemesis
Epistaxis

Increased sexual desire
Sexual indiscretions
Violent sex

Red tongue with yellow coating
Strong, Wiry & Rapid Pulse

Prognosis & Complication

Liver Yin Xu & Liver Blood Xu
Stomach Fire
Heart Fire

Differentiation

Liver Fire vs. Stomach Fire vs. Heart Fire
Liver Qi vs. Liver Heat vs. Liver Fire

Therapeutics:

The treatment principle is to clear the Liver and purge the Liver fire. This is a reducing method. It aims to make the Liver clear and pass heat and fire from urine and stool.

Acupuncture:

Xing Jian	LV 2
Feng Chi	GB 20
Zan Zhu	UB 2
Tai Yang	Extra

Herbal Formulation:

Long Dan Xie Gan Tang 龙胆泻肝汤

- | | |
|----------------------------------|------|
| • Long Dan Cao (gentiana) | 9gm |
| • Huang Qin (Scutellaria) | 9gm |
| • Zhi Zi (gasdenia fruit) | 9gm |
| • Mu tong (Akebia) | 6gm |
| • Sheng Di Huang (raw rehmannia) | 15gm |
| • Chen Qian Zi (plantago seed) | 9gm |
| • Ze Xie (alisma) | 9gm |
| • Dang Gui (angelica) | 9gm |

- Chai Hu (bupleurum) 9gm
- Gan Cao (licorice) 3gm

3. Deficiency of Liver Blood

This pattern is due to hemopoiesic impairment caused by deficiency of the Spleen and the Kidney which are the main sources of the blood. The other factor is chronic diseases that will injure the blood. Prolonged depression of Liver Qi may consume the Liver blood and loss of blood also damages the function of Liver storing blood and vice versa.

Clinical manifestation:

1. Dry eyes
2. Hypochondriac pain
3. General blood Xu

Pale conjunctivas

Blurry vision

Flowery vision

Bird Vision

Weakness in the limbs,

Numbness in muscles,

Rigidity and stiffness in the joints,

Tremors,

Nails are also dry and brittle, they are the surplus of tendon.

Beauty of hair is diminished, thin, dry and brittle, and falls out, no color changes

Scanty light-colored menstrual flow

Amenorrhea

Infertility

Men's Prostatic fluid, they do not have blood supply to prostate therefore they have ejaculation problem

Small, narrow and pale tongue

Thready or wiry & thready pulse

Prognosis & Complication

Heart Blood Xu and General Blood Xu

Liver Yin Xu

Kidney Essence Xu

Differentiation

Liver Blood xu vs. Heart Blood xu

Therapeutics:

The treatment principle is to nourish the blood and strengthen the Liver's function of storing blood.

Acupuncture:

Zu San Li	ST 36
Pi Shu	UB 20
Tai Chong	LV 3
Gan Shu	UB 18
Guang Ming	GB 37
San Yin Jiao	SP 6

Herbal Formulation:

Bu Gan Tang 补肝汤

- | | |
|---------------------------------|------|
| • Dang Gui (angelica) | 9gm |
| • Bai Shao (white peony) | 15gm |
| • Chuan Xiong (ligusticum) | 9gm |
| • Shen Di Huang (raw rehmannia) | 15gm |
| • Suan Zao Ren (ziziphus) | 9gm |
| • Mu Gua (chaenomeles fruit) | 9gm |
| • Zhi Gan Cao (licorice) | 3gm |

4. Insufficiency of Liver Yin

This reflects an impairment and deficiency of Liver Yin, which is unable to control Yang, leading to deficiency-type internal heat. These are cellular changes that may become permanent. This is due to emotional disharmonies causes depression of Liver Qi which transforms into Fire, or Liver disorders and febrile disease which injures Liver Yin.

Clinical Manifestation:

1. Impaired eyesight
2. Hypochondriac pain
3. Deficient heat

Dry eyes

Blurry vision

Flying flies

Floater

Halos around lights

Blind spot

Hypochondriac pain, most on right side of hypochondriac area

Dizziness

Flush of the face

Feverish sensation in the palms, soles, and epigastric area

Afternoon fever

Night sweating

Dry throat
Dry skin
Constipation

Small dry red tongue, no coating or thin coating or geographic coating
Wiry, thready and rapid pulse

Prognosis & Complication

Kidney Yin Xu
Liver Yang Hyperactivity

Differentiation

Liver Blood vs. Liver Yin Xu

Therapeutics:

The treatment principle is to nourish Liver Yin and reduce the deficient heat. Since Liver Yin and Kidney essence are mutually engendering, treatment usually involves the Liver and Kidney together.

Acupuncture:

Gan Shu	UB 18
Shen Shu	UB 23
Tai Xi	KD 3
Qu Quan	LV 8
San Yin Jiao	SP 6

Herbal Formulation:

Yi Guang Jian 一贯煎

- | | |
|---------------------------------|------|
| • Gou Qi Zi (lycium) | 9gm |
| • Shen Di Huang (raw rehmannia) | 15gm |
| • Sha Shen (glehnia) | 12gm |
| • Mai Meng Dong (Ophiopogon) | 9gm |
| • Dang Gui (angelica) | 9gm |
| • Chuan Lian Zi (chinaberry) | 6gm |

5. Hyperactivity of Liver Yang

This pattern is the manifestation of water not nourishing and controlling the wood. Under this condition, Liver Yang becomes hyperactive, it manifests as upper excess and lower deficiency. It is mainly due to Liver Yin and Kidney Yin deficiency failing to control Liver Yang. Anger and anxiety that cause stagnation of the Qi and causes Liver Fire will consume the Yin and fluid causing an imbalance of Kidney Yin and Liver Yang.

Clinical Manifestation:

1. Irritability
2. Dizziness

3. Lower back pain and weakness of knees

Dizziness

Headache

Distension and pain in the eyes

Flushed complexion

Red eyes

Irritability

Anger

Palpitation

Insomnia

Excessive dreaming

Lower back pain

Weakness of knees

High sexual desire

Poor (Dry) sexual function

Red tongue; yellow coating, or no coating

Wiry and strong or thready, and rapid pulse.

Prognosis & Complication

Liver Wind

Differentiation

Liver Heat

Liver Fire

Liver Yang

Therapeutics:

The treatment principle is to subdue the Liver Yang and nourish the Liver and Kidney Yin.

Acupuncture:

Tai Chong LV 3

Xin Jian LV 2

Feng Chi GB 20

Bai HUI DU 20

Herbal Formulation:

Tian Ma Gou Teng Yin 天麻钩藤饮

- Tian Ma (gastrodia tuber) 9gm
- Gou Teng (uncaria) 9gm
- Shi Jue Ming (sea-ear Shell) 9gm
- Sang Ji Sheng (mulberry mistletoe) 9gm
- Du Zhong (eucommia bark) 9gm

• Chuan Niu Xi (cyathula root)	9gm
• Zhi Zi (gardenia fruit)	6gm
• Huang Qin (scutellaria)	9gm
• Yin Mu Cao (leonurus)	9gm
• Ye Jiao Teng (fleeceflower)	9gm
• Fu Ling (poria)	9gm

6. Stirring of Liver Wind

The concept of the Wind in Chinese medicine indicates rapid changes and constant movement. The characteristics of Liver Wind are dizziness, vertigo, convulsion, tremor and the other “shaking” symptoms and signs. Generally there exist four kinds of stirring of Liver Wind: utmost Heat producing Wind; hyperactivity of Liver Yang producing Wind; deficient Yin producing Wind, and deficient Blood producing Wind. Here we only discuss the hyperactivity of the Liver Yang producing wind because it is the most common and most important in the clinical practice. It is due to deficiency of the Liver and Kidney Yin. Liver Yang rebels and rises suddenly, it stirs up internal Liver Wind.

Clinical Manifestation:

1. Dizziness
2. Shaking
3. Tremor
4. Sudden Fall
5. Lost consciousness

Dizziness and vertigo

Headache with shaking movement

Rigidity and stiffness in the neck

Tremor in the limbs

Numbness in the hand and foot.

Speech defect

Sudden fall

Coma

Gurgling with sputum in the throat

Hemiplegia

Deviation of the mouth and eyes

Red tongue, thick yellow sticky coating

Strong wiry pulse

Prognosis & Complication

Death

Paralysis

Multiple organ impairments

Differentiation

Liver Qi stagnation
Liver Heat Phenomena
Liver Fire Uup-flaming
Liver Yang Hyperactivity
Liver Wind Stirring

Hyperactivity of Liver Yang causing Wind
Extreme Heat generating Wind
Deficient Yin causing Wind
Deficient Blood causing Wind

Therapeutics:

The treatment principle is to calm the Liver and extinguish internal Wind at its root by nourishing Yin and subduing Yang.

Acupuncture:

Feng Chi	GB 20
Tai Chong	LV 3
Feng Fu	DU 16
Da Zhui	DU 14
He Gu	LI 4
Yong Quan	KD 1
Ren Zhong	DU 26

Herbal Formulation:

EXTINGUISHING LIVER WIND FORMULA

Zhen Gan Xi Feng Tang 镇肝熄风汤

- | | |
|----------------------------------|------|
| • Huan Niu Xi (achyranthes) | 15gm |
| • Xuan Shen (scrophularia) | 12gm |
| • Tian Men Dong (asparagus) | 9gm |
| • Bai Shao (white peony) | 12gm |
| • Yin Chen Hao (capillaria) | 6gm |
| • Chuan Lian Zi (chinaberry) | 6gm |
| • Mai Ya (malt) | 6gm |
| • Gan Cao (licorice) | 6gm |
| • Dai Zhe Shi (hematite) | 15gm |
| • Long Gu (dragon's bone) | 15gm |
| • Mu Li (oystose shell) | 15gm |
| • Gui Ban (tortose Plastron) | 15gm |
| • Sheng Di Huang (raw rehmannia) | 15gm |

7. Stagnation of Cold in the Liver Channel

This pattern reflects the accumulation of pathogenic cold in the Liver channel, presenting as cold-type pain along the course of the channel. It is caused by exogenous pathogenic Cold invading the body and depression of the Liver Qi.

Clinical Manifestation:

1. Lower abdominal cramping, most commonly seen in males.
2. Referring pain to testicles and scrotum.

Prolapse-like sensation.

Pain alleviated by warmth.

Pain aggravated by cold.

Vertex headache (Cold sensation)

Thick white smooth coating, pale tongue.

Deep, wiry or tight, slow pulse.

Prognosis & Complication

Local Qi and Blood stagnation

Spleen Qi and Yang Xu

Infertility

Differentiation

Cold Stagnation the Liver Channel vs. Small Intestine Qi Stagnation (Inguinal Hernia)

Cold Stagnation the Liver Channel vs. Cold in the Blood

Therapeutics:

The treatment principle is to invigorate circulation of Qi and disperse the pathogenic Cold.

Acupuncture:

Da Dun	LV 1
Tai Chong	LV 3
Qi Hai	Ren 6
Guan Yuan	Ren 4
Gui Lan	ST 29

Herbal Formulation:

Nuan Gan Jian 暖肝煎

- Dang Gui (angelica) 9gm
- Gou Qi Zi (lycium) 9gm
- Xiao Hui Xiang (fennel fruit) 9gm
- Rou Gui (cinnamon bark) 6gm
- Wu Yao (lindera) 15gm
- Chen Xiang (aquilaria) 3gm
- Fu Ling (poria) 12gm
- Sheng Jiang (Ginger) 6gm

8. Damp Heat in the Liver and Gallbladder

This pattern is the manifestation of Damp Heat accumulating in the Liver and Gallbladder, impairs the smooth Qi flow and normal organ function. This is caused by pathogenic Damp Heat invades the body; eating too much greasy and sweet food; or dysfunction of the Spleen and Stomach leads to internal dampness, which gradually accumulates and transforms into heat.

Clinical Manifestation:

1. Hypochondriac pain and distension.
2. Jaundice

Damp heat in the liver:

Hypochondriac pain and distension (Right side)

Liver enlargement

Yellow eye

Yellow skin

Yellow urination

Low-grade fever

Joint pain

Loss of appetite

Abdominal distension

Irregular bowel movement

Some diarrhea

Damp heat in the Gall bladder:

Alternating fever and chills.

Yellow Sclera and mucous membranes

Abdominal cramping referring right shoulder.

Murphy's sign during physical examination (breathe in).

Bitter taste

Nausea

Irregular bowel movement.

Some constipation with pale stools.

1. No Jaundice!!!
2. The liver channel symptoms and signs

No Jaundice!!!

Eczema in the scrotum with heavy itching.

Pain and distension in the testes.

Leucorrhea with yellow color and fetid odor.

Vaginal itching.

Red tongue with Yellow Sticky coating
Wiry, slippery and rapid pulse

Prognosis & Complication

Damp Heat in the Liver Channel
Damp Heat in the Liver and Gall bladder
Qi or Blood Stagnation
Liver Yin Xu
Spleen Qi Xu

Differentiation

Damp Heat in Liver channel vs. Cold Stagnation in Liver Channel
Damp Heat in Liver channel vs. Damp Heat in Liver organ
Differentiation of jaundice

Therapeutics:

The treatment principle is to regulate the Liver function and eliminate Damp Heat.

Acupuncture:

Gan Shu	UB 18
Zi Yang	DU 9
Wan Gu	SI 4
Yang Ling Quan	GB 34
Tan Chong	LV 3

Herbal Formulation:

Yin Chen Hao Tang 茵陈蒿汤

- Yin Chen Hao (capillaris) 18gm
- Zhi Zi (gardenia fruit) 9gm
- Da Huang (rhubarb) 6gm

9. Depression of the Gallbladder and Disturbance of Phlegm

This pattern reflects the failure of the Gallbladder's function of excretion and discharge. Internally, phlegm-heat disturbs the system and disrupts the smooth flow of Liver Qi. Emotional frustration or depression can lead to Liver Qi stagnation, which may transform into Fire, consuming fluids and generating Phlegm.

Clinical Manifestation:

1. Cowardice.
2. Insomnia.
3. Nausea.
4. Yellow belly (Eng)
5. Small Gall Bladder (Chin)

Lack of courage.

Palpitation.
Cowardice.
Restlessness.
Insomnia.
Dream disturbed sleep.
Can not make a decision.

Hypochondriac pain.
Chest stiffness.
Bitter taste.
Sighing.
Nausea.
Dry vomiting.
Dizziness and vertigo.

Yellow greasy tongue coating
Wiry, slippery, and rapid pulse.

Prognosis & Complication

Liver Qi stagnation
Heart Shen disturbed
Spleen and Stomach dysfunction

Differentiation

Gall Bladder Phlegm Depression
Liver Qi Stagnation

Therapeutics:

The treatment principle is to regulate Qi and transform Phlegm; harmonize the relationships between the Gallbladder and the Heart, and between the Gallbladder and the Stomach.

Acupuncture:

Wei Shu	UB 21
Zu San Li	ST 36
Feng Long	ST 40
Dan Shu	UB 19
Yin Lin Quan	SP 9
San Yin Jiao	SP 6

Herbal Formulation:

Huang Lian Wen Dan Tang 黄连温胆汤

- | | |
|-------------------------|------|
| • Huang Lian (coptis) | 6gm |
| • Bai Xia (pinellia) | 6gm |
| • Chen Pi (citrus peel) | 9gm |
| • Fu Ling (poria) | 12gm |

- Zhi Shi (immature bitter orange) 9gm
- Zhu Ru (bamboo shavings) 9gm
- Zhi Gan Cao (licorice) 6gm
- Da Zao (jujube) 5pc
- Sheng Jiang (Ginger) 3pc